COUNTDOWN CALCULATOR

Title of the Project

Countdown Timer

Aim

The Countdown Timer App provides users a simple and efficient way to run a countdown timer based on hours, minutes, and seconds input, using Python Tkinter with a user-friendly UI.

Explanation

Introduction:
The Countdown Timer is a Python desktop application built using Tkinter. Users can input hours, minutes, and seconds, and start a timer which counts down to zero. Once the time is up, a message is shown.

Project Setup:
- Install Python 3.x
- Tkinter comes pre-installed with Python.
- Run the .py file to launch the GUI application.

Dependencies Used:
- Tkinter library for GUI development.

UI Design

The application window includes:
- Entry fields for Hours, Minutes, and Seconds.
- A Start Timer button.
- A Label to show the countdown time.

Colors:

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- Background color (#FAF3E0)
- Frame and Button styled with custom colors for better appearance.

Program

```
import tkinter as tk from
tkinter import messagebox

def initiate_timer():
try:
        hrs = int(hours_input.get())
mins = int(minutes_input.get())
secs = int(seconds_input.get())
except ValueError:
        messagebox.showerror("Input Error", "Enter valid numbers
only!")         return

    time_left = hrs * 3600 + mins * 60 + secs

    start_btn.config(state="disabled")  # Disable start button once started

    def update_timer():
nonlocal time_left
if time_left >= 0:
            display_hrs, remainder = divmod(time_left, 3600)
display_mins, display_secs = divmod(remainder, 60)

countdown_label.config(text=f"{display_hrs:02d}:{display_mins:02d}:{display_secs
:02d}")             time_left -= 1
            root.after(1000, update_timer)
else:
            messagebox.showinfo("Timer Finished", "Time's up!")
start_btn.config(state="normal")  # Enable button again

    update_timer()

# Create main application window
root = tk.Tk()
root.title("Simple Countdown Timer")
root.geometry("420x420")
root.configure(bg="#FAF3E0")
```

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```
container = tk.Frame(root, bg="#F7C59F", bd=4, relief="groove")
container.place(relx=0.5, rely=0.5, anchor="center")

header = tk.Label(container, text="Countdown Timer", font=("Verdana", 22,
"bold"), bg="#F7C59F") header.

tainer, text="00:00:00", font=("Verdana", 28), bg="#F7C59F")
countdown_label.pack(pady=10)

input_section = tk.Frame(container, bg="#F7C59F")
input_section.pack(pady=10)

hours_input = tk.Entry(input_section, width=4, font=("Verdana", 18))
hours_input.grid(row=0, column=0, padx=5)
minutes_input = tk.Entry(input_section, width=4, font=("Verdana", 18))
minutes_input.grid(row=0, column=1, padx=5)
seconds_input = tk.Entry(input_section, width=4, font=("Verdana", 18))
seconds_input.grid(row=0, column=2, padx=5)

start_btn = tk.Button(container, text="Start Timer", font=("Verdana", 16),
bg="#EC6E4C", fg="white", command=initiate_timer) start_btn.pack(pady=20)

root.mainloop()
```

Output

After entering the time and clicking Start Timer:

- The timer will display the countdown in hh:mm:ss format.

- When the countdown finishes, a pop-up 'Time's up!' message appears.

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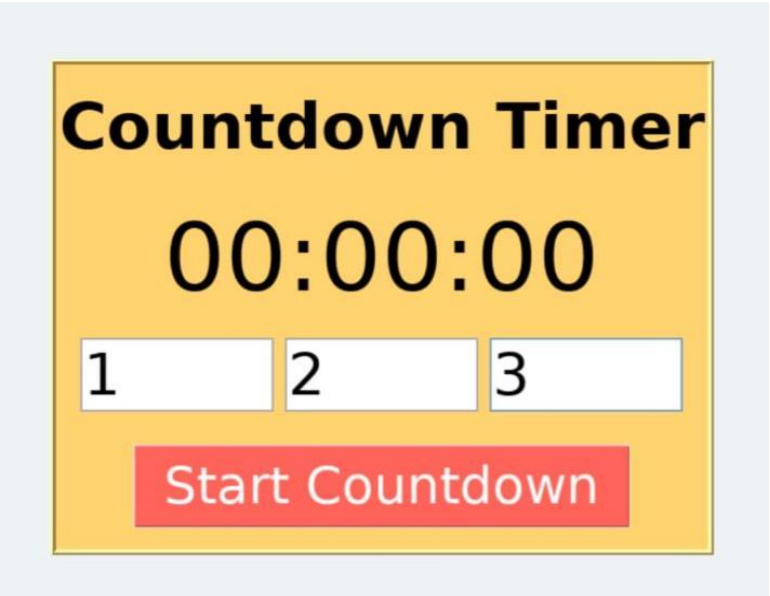

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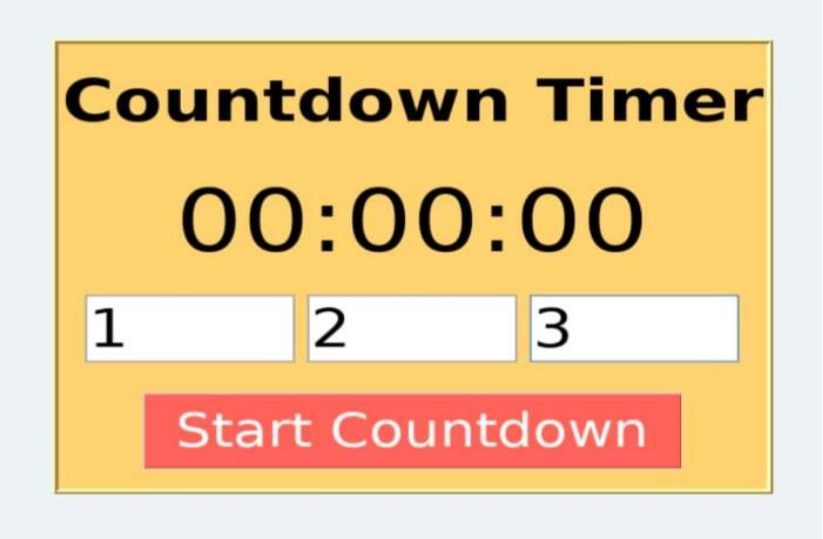

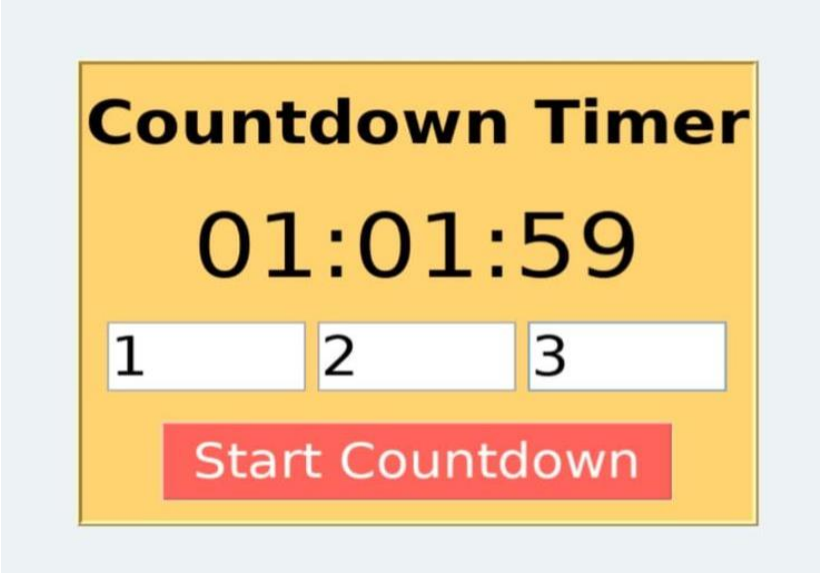

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